Envelope Equations

USPAS - Beam Physics with Intense Space Charge

Austin Hoover   Steven Lund   Daniel Winklehner   John Barnard

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Outline I

1. Paraxial ray equation

2. Envelope equations for axially symmetric beams

3. Envelope equations for elliptically symmetric beams

0.1 Envelope equations: reduced-order characterization of beam dynamics in terms of second-order moments

[Image]

0.2 Road map

- Start from single particle equation of motion: $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.
- Make use of:
  - Paraxial (near-axis) approximation
  - Conservation of angular momentum
  - Axisymmetry of external forces
- Arrive at single-particle paraxial ray equation.
- Take statistical averages to obtain moment/envelope equations.

Outline

1. Paraxial ray equation
2. Envelope equations for axially symmetric beams
3. Envelope equations for elliptically symmetric beams

1.1 Equation of motion in cylindrical coordinates

Equation of motion for particle of mass m and charge q in electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$:

$$\frac{d\mathbf{p}}{dt} = \frac{d(\gamma m \dot{\mathbf{x}})}{dt} = q\left(\mathbf{E} + \dot{\mathbf{x}} \times \mathbf{B}\right), \tag{1}$$

Switch to cylindrical coordinates (r, θ, z):

$$\begin{aligned} \mathbf{p} &= p_x \hat{x} + p_y \hat{y} + p_z \hat{z} \\ &= p_r \hat{r} + p_\theta^* \hat{\theta} + p_z \hat{z}. \end{aligned} \tag{2}$$

The components p_r and p_θ^* are given by

$$\begin{aligned} p_r &= \gamma m \dot{r}, \\ p_\theta^* &= \gamma m r \dot{\theta}. \end{aligned} \tag{3}$$

The symbol p_θ^* is used to reserve p_θ for the *canonical* angular momentum, defined later.

In cylindrical coordinates, the unit vectors depend on the particle position and thus change with time:

$$\begin{aligned} \dot{\hat{r}} &\equiv d\hat{r}/dt = \dot{\theta}\hat{\theta}, \\ \dot{\hat{\theta}} &\equiv d\hat{\theta}/dt = -\dot{\theta}\hat{r}. \end{aligned} \tag{4}$$

Thus, the time derivative of the momentum vector in Eq. (2) is:

$$\begin{aligned}
\frac{d\mathbf{p}}{dt} &= \frac{d}{dt}\left(p_r\hat{r} + p_\theta^*\hat{\theta} + p_z\hat{z}\right) && (5) \\
&= \dot{p}_r\hat{r} + p_r\dot{\hat{r}} + \dot{p}_\theta\hat{\theta} + p_\theta^*\dot{\hat{\theta}} + \dot{p}_z\hat{z} && (6) \\
&= \left(\dot{p}_r - p_\theta^*\dot{\theta}\right)\hat{r} + \left(\dot{p}_\theta + p_r\dot{\theta}\right)\hat{\theta} + \dot{p}_z\hat{z} && (7) \\
&= \left[\frac{d(\gamma m\dot{r})}{dt} - \gamma m r\dot{\theta}^2\right]\hat{r} + \left[\frac{d(\gamma m r\dot{\theta})}{dt} + \gamma m\dot{r}\dot{\theta}\right]\hat{\theta} + \left[\frac{d(\gamma m\dot{z})}{dt}\right]\hat{z}. && (8) \\
&= \left[\frac{d(\gamma m\dot{r})}{dt} - \gamma m r\dot{\theta}^2\right]\hat{r} + \left[\frac{1}{r}\frac{d}{dt}\left(\gamma m r^2\dot{\theta}\right)\right]\hat{\theta} + \left[\frac{d(\gamma m\dot{z})}{dt}\right]\hat{z}. && (9)
\end{aligned}$$

This takes care of the left side of the Lorentz force equation (Eq. (1)).

The right side of Eq. (1) requires the cylindrical field components:

$$\begin{aligned}\mathbf{E} &= E_r\hat{r} + E_\theta\hat{\theta} + E_z\hat{z},\\ \mathbf{B} &= B_r\hat{r} + B_\theta\hat{\theta} + B_z\hat{z}.\end{aligned} \tag{10}$$

Evaluating the cross product $\dot{\mathbf{x}} \times \mathbf{B}$,

$$\dot{\mathbf{x}} \times \mathbf{B} = \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{z} \\ \dot{r} & r\dot{\theta} & \dot{z} \\ B_r & B_\theta & B_z \end{vmatrix}, \tag{11}$$

leads to

$$\frac{d(\gamma m\dot{r})}{dt} - \gamma m r\dot{\theta}^2 = q\left(E_r + r\dot{\theta}B_z - \dot{z}B_\theta\right), \tag{12}$$

$$\frac{1}{r}\frac{d(\gamma m r^2\dot{\theta})}{dt} = q\left(E_\theta + \dot{z}B_r - \dot{r}B_z\right), \tag{13}$$

$$\frac{d(\gamma m\dot{z})}{dt} = q\left(E_z + \dot{r}B_\theta - r\dot{\theta}B_r\right) \tag{14}$$

1.2 From fields to potentials

Write the fields in terms of the scalar potential $\phi(\mathbf{x})$ and vector potential $\mathbf{A}(\mathbf{x})$:

$$\begin{aligned} \mathbf{E} &= -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}, \\ \mathbf{B} &= \nabla \times \mathbf{A}. \end{aligned} \tag{15}$$

The gradient $\nabla\phi$ in cylindrical coordinates is:

$$\nabla\phi = \left[\frac{\partial\phi}{\partial r}\right]\hat{r} + \left[\frac{1}{r}\frac{\partial\phi}{\partial\theta}\right]\hat{\theta} + \left[\frac{\partial\phi}{\partial z}\right]\hat{z}. \tag{16}$$

And the curl $\nabla \times \mathbf{A}$ is:

$$\nabla \times \mathbf{A} = \left[\frac{1}{r}\frac{\partial A_z}{\partial\theta} - \frac{\partial A_\theta}{\partial z}\right]\hat{r} + \left[\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}\right]\hat{\theta} + \frac{1}{r}\left[\frac{\partial(rA_\theta)}{\partial r} - \frac{\partial A_r}{\partial\theta}\right]\hat{z}. \tag{17}$$

Axial symmetry means all partial derivatives with respect to θ are zero in Eq. (16) and Eq. (17). This gives the following expressions for the fields in terms of the potentials in cylindrical coordinates:

$$\begin{aligned} \mathbf{E} &= \left[-\frac{\partial \phi}{\partial r} - \frac{\partial A_r}{\partial t}\right]\hat{r} + \left[-\frac{\partial A_\theta}{\partial t}\right]\hat{\theta} + \left[-\frac{\partial \phi}{\partial z} - \frac{\partial A_z}{\partial t}\right]\hat{z}, \\ \mathbf{B} &= \left[-\frac{\partial A_\theta}{\partial r}\right]\hat{r} + \left[\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}\right]\hat{\theta} + \left[\frac{1}{r}\frac{\partial (r A_\theta)}{\partial r}\right]\hat{z}. \end{aligned} \tag{18}$$

1.3 Conservation of angular momentum

Use the potentials to rewrite the angular equation of motion (Eq. (13)):

$$
\begin{aligned}
\frac{d}{dt}\left(\gamma m r^2 \dot{\theta}\right) &= qr\left(E_\theta + \dot{z}B_r - \dot{r}B_z\right) \\
&= qr\left(\left[-\frac{\partial A_\theta}{\partial t}\right] + \dot{z}\left[-\frac{\partial A_\theta}{\partial z}\right] - \dot{r}\left[\frac{1}{r}\frac{\partial(rA_\theta)}{\partial r}\right]\right) \\
&= -q\left(\frac{\partial(rA_\theta)}{\partial t} + \dot{z}\frac{\partial(rA_\theta)}{\partial z} + \dot{r}\frac{\partial(rA_\theta)}{\partial r}\right) \\
&= -q\left[\frac{\partial}{\partial t} + \dot{z}\frac{\partial}{\partial z} + \dot{r}\frac{\partial}{\partial r}\right](rA_\theta) \\
&= -q\left[\frac{\partial}{\partial t} + \dot{\mathbf{x}}\cdot\nabla\right](rA_\theta) \\
&= -\frac{d(qrA_\theta)}{dt}.
\end{aligned} \tag{19}
$$

Eq. (19) can then be written as:

$$\frac{d}{dt}\left(\gamma m r^2 \dot{\theta} + q r A_\theta\right) = 0. \tag{20}$$

This motivates the definition of the *canonical angular momentum*:

$$p_\theta \equiv \gamma m r^2 \dot{\theta} + q r A_\theta, \tag{21}$$

such that $dp_\theta/dt = 0$. We may also write p_θ in terms of the magnetic flux $\psi(r)$ though a circle of radius r *[Figure]*:

$$\psi(r) = \int \mathbf{B} \cdot d\Omega = \int \nabla \times \mathbf{A} \cdot d\Omega = \oint \mathbf{A} \cdot d\mathbf{l} = 2\pi r A_\theta. \tag{22}$$

Therefore, Eq. (21) can be written

$$p_\theta \equiv \gamma m r^2 \dot{\theta} + \frac{q\psi(r)}{2\pi}. \tag{23}$$

1.4 Series expansion of axially symmetric fields

Consider external fields $\mathbf{E}$ and $\mathbf{B}$ with azimuthal symmetry ($\partial/\partial\theta = 0$). In the absence of charge and current densities,

$$\begin{aligned} \nabla \cdot \mathbf{E} &= 0, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= 0, \\ \nabla \times \mathbf{B} &= 0. \end{aligned} \tag{24}$$

The vanishing curl means we can express the fields as

$$\begin{aligned} \mathbf{E} &= -\nabla\phi, \\ \mathbf{B} &= -\nabla\phi_m, \end{aligned} \tag{25}$$

for scalar potentials ϕ and ϕ_m. Additionally, the vanishing divergence leads to the Laplace equation for the scalar potential:

$$\nabla^2\phi = 0. \tag{26}$$

In cylindrical coordinates, the Laplace equation for the potential is:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \phi}{\partial r}\right)+\frac{\partial^2 \phi}{\partial z^2}=0. \tag{27}$$

The azimuthal symmetry will lead to a simplified form of the scalar potential in terms of its on-axis derivatives along the z axis. First, expand the potential $\phi(r, z)$ as a power series,

$$\phi(r, z)=\sum_{n=0}^{\infty} h_{2n}(z) r^{2n}, \tag{28}$$

where the $h_{2n}(z)$ are functions of z. The on-axis potential is given by $h_0(z)=\phi(0, z)$.

The rotational symmetry eliminates all odd powers of r in the series.

At the origin ($r = 0$), we must have $E_r = B_r = 0$. If we allowed the $n = 1$ term $h_1(z)r$, we would have $E_r(0, z) = -\partial\phi(r, z)/\partial r|_{r=0} = h_1(z) \neq 0$.

The derivatives in the Laplace equation are:

$$\begin{aligned} \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \phi}{\partial r}\right) &= \sum_{n=1}^{\infty}(2n)^2 h_{2n}(z) r^{2n-2}, \\ \frac{\partial^2 \phi}{\partial z^2} &= \sum_{n=0}^{\infty} h''_{2n}(z) r^{2n}, \end{aligned} \tag{29}$$

where $h'(z) = \partial h(z)/\partial z$.

Thus, $\nabla^2\phi = 0$ becomes:

$$\begin{aligned}
\sum_{n=1}^{\infty}(2n)^2 h_{2n}(z) r^{2n-2} + \sum_{n=0}^{\infty} h''_{2n}(z) r^{2n} &= 0 \\
\sum_{n=0}^{\infty}(2n+2)^2 h_{2n+2}(z) r^{2n} + \sum_{n=0}^{\infty} h''_{2n}(z) r^{2n} &= 0 \\
\sum_{n=0}^{\infty}\left[(2n+2)^2 h_{2n+2}(z) + h''_{2n}(z)\right] &= 0
\end{aligned} \tag{30}$$

The result is a recurrence relation,

$$h_{2n+2}(z) = -\frac{h''_{2n}(z)}{(2n+2)^2}, \tag{31}$$

with which we can write all terms as derivatives of $h_0(z)$ with respect to z:

$$\begin{aligned}
h_2(z) &= -\frac{1}{4}\frac{\partial^2 h_0(z)}{\partial z^2}, \\
h_4(z) &= +\frac{1}{64}\frac{\partial^4 h_0(z)}{\partial z^4}, \\
&\vdots
\end{aligned} \tag{32}$$

We may thus write the potential at all points in space using only the z-derivatives of the on-axis potential $\phi(0, z)$:

$$\phi(r, z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \frac{\partial^{2n} \phi(0, z)}{\partial z^{2n}} \left(\frac{r}{2}\right)^{2n} . \tag{33}$$

We now express the fields in terms of the expanded scalar potentials. Write the magnetic field as $\mathbf{B} = B_r \hat{r} + B_z \hat{z}$. Using the series expansion of $\phi_m(r,z)$, we find for the z component of the field:

$$
\begin{aligned}
B_z(r,z) &= -\frac{\partial \phi_m}{\partial z} \\
&= -\frac{\partial}{\partial z}\left[\sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n}\phi_m(0,z)}{\partial z^{2n}} \left(\frac{r}{2}\right)^2\right] \\
&= -\sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n+1}\phi_m(0,z)}{\partial z^{2n+1}} \left(\frac{r}{2}\right)^2 .
\end{aligned}
\tag{34}
$$

Define $B(z)$ as the on-axis value of $B_z(z)$:

$$
B(z) \equiv B_z(0,z) = -\frac{\partial \phi_m(0,z)}{\partial z}. \tag{35}
$$

Inserting Eq. (35) into Eq. (34) gives:

$$
\begin{aligned}
B_z(r,z) &= \sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n}}{\partial z^{2n}} \left[-\frac{\partial \phi_m(0,z)}{\partial z} \right] \left(\frac{r}{2}\right)^2 \\
B_z(r,z) &= \sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n} B(z)}{\partial z^{2n}} \left(\frac{r}{2}\right)^2 .
\end{aligned} \tag{36}
$$

For the radial component, we find:

$$
B_r(r,z) = \sum_{n=1}^{\infty} \frac{(-1)^{n}}{n!(n-1)!} \frac{\partial^{2n-1} B(z)}{\partial z^{2n-1}} \left(\frac{r}{2}\right)^{2n-1} . \tag{37}
$$

A similar procedure gives the radial and longitudinal components of the electric field. Define $V(z)$ as the on-axis value of the electric potential ϕ:

$$V(z) \equiv \phi(0, z). \tag{38}$$

This gives:

$$\begin{aligned} E_z(r, z) &= -\frac{\partial \phi}{\partial z} = -\sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \frac{\partial^{2n+1} V(z)}{\partial z^{2n+1}} \left(\frac{r}{2}\right)^{2n}. \\ E_r(r, z) &= -\frac{\partial \phi}{\partial r} = -\sum_{n=1}^{\infty} \frac{(-1)^n}{n!(n-1)!} \frac{\partial^{2n} V(z)}{\partial z^{2n}} \left(\frac{r}{2}\right)^{2n-1}. \end{aligned} \tag{39}$$

1.5 Derivation of paraxial ray equation

We will now take the *paraxial* limit, assuming small transverse position and velocity: $\dot{r} \ll \dot{z}$, $r\dot{\theta} \ll \dot{z}$. If $v = (\dot{r}^2 + r^2\dot{\theta}^2) + \dot{z}^2)^{1/2}$ is the total velocity, the paraxial limit gives $\dot{z} = (v^2 - \dot{r}^2 - r^2\dot{\theta}^2)^{1/2} \approx v$.

To first order in r and r', the external fields are:

$$\begin{aligned} E_r^{(ext)} &\approx \frac{1}{2}\frac{\partial^2 V(z)}{\partial z^2} r, \\ B_z^{(ext)} &\approx B(z), \\ B_\theta^{(ext)} &= 0 \quad (\text{since } \partial\phi_m/\partial\theta = 0). \end{aligned} \tag{40}$$

Plug these truncated fields into the radial equation of motion (Eq. (12)):

$$\begin{aligned} \frac{d(\gamma m \dot{r})}{dt} - \gamma m r\dot{\theta}^2 &= q\left(E_r + r\dot{\theta}B_z - \dot{z}B_\theta\right), \\ &\approx q\left(\frac{1}{2}\frac{\partial^2 V(z)}{\partial z^2} r + r\dot{\theta}B(z)\right) \end{aligned} \tag{41}$$

Use z as the independent variable: $dz = \beta c dt$. Using $' \equiv d/dz$ and dropping the $\approx$ symbol gives:

$$\beta c \frac{d\left(\gamma m \beta c r'\right)}{dz} - \gamma m r (\beta c \theta')^2 = q\left(\frac{1}{2} V''(z) r + r \beta c \theta' B(z)\right). \tag{42}$$

Expand the derivative in the first term and divide by $\gamma \beta^2 m c^2$:

$$r'' + \frac{(\gamma\beta)'}{(\gamma\beta)} r' - r\theta'^2 = \frac{q}{\gamma \beta^2 m c^2}\left(\frac{1}{2} V''(z) + \beta c \theta' B(z)\right) r. \tag{43}$$

Now use the definition of the canonical angular momentum p_θ (Eq. (21)) to eliminate θ', noting that the flux $\psi(r) = \pi r^2 B$:

$$\begin{aligned} \theta' &= \frac{\dot{\theta}}{\beta c} \\ &= \frac{1}{\beta c}\frac{1}{\gamma m r^2}\left(p_\theta - \frac{q\psi(r)}{2\pi}\right), \\ &= \frac{p_\theta}{\gamma m \beta c r^2} - \frac{\omega_c}{2\gamma\beta c}. \end{aligned} \tag{44}$$

We have defined the *cyclotron frequency* $\omega_c \equiv \frac{qB}{m}$.

Substitute Eq. (44) into Eq. (43):

$$r'' + \frac{(\gamma\beta)'}{\gamma\beta} r + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2 r - \left(\frac{p_\theta}{\gamma m \beta c}\right)^2 \frac{1}{r^3} = \frac{q}{\gamma m \beta^2 c^2}\frac{1}{2}V''(z) r \tag{45}$$

The rate of change in the particle energy is given by

$$\frac{d(\gamma mc^2)}{dt} = q\mathbf{E} \cdot \mathbf{v}. \tag{46}$$

Rearrange and switch from t to z. In the paraxial limit:

$$\begin{aligned}
\gamma' &= \frac{q}{mc^2}\frac{\mathbf{E} \cdot \mathbf{v}}{v_z} && (47)\\
&= \frac{q}{mc^2}\frac{E_x v_x + E_y v_y + E_z v_z}{v_z} && (48)\\
&\approx \frac{q}{mc^2} E_z. && (49)
\end{aligned}$$

Giving the approximate relationship:

$$\gamma'' \approx \frac{q}{mc^2} E_z' = -\frac{q}{mc^2} V''. \tag{50}$$

This approximation of λ'' leads to the *paraxial ray equation*:

$$\boxed{r'' + \frac{(\gamma\beta)'}{(\gamma\beta)} r' + \left(\frac{\gamma''}{2\gamma\beta^2}\right) r + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2 r - \left(\frac{p_\theta}{\gamma m \beta c}\right)^2 \frac{1}{r^3} = 0.} \qquad (51)$$

Terms in order: (i) inertial term; (ii) accelerative damping; (iii) E_r from converging field lines; (iv) solenoidal focusing ($v_\theta B_z$, part of centrifugal term); (v) centrifugal defocusing.

1.6 Addition of self-generated fields

We can also account for linearized self-fields $\mathbf{E}^{(self)}$ and $\mathbf{B}^{(self)}$. Adding these self-generated fields to the radial equation of motion gives:

$$r'' + \frac{(\gamma\beta)'}{(\gamma\beta)} - r\theta'^2 = \frac{q}{mc^2\gamma\beta^2}\left(\frac{1}{2}V''(z)r + \beta c\theta' B(z)r + E_r^{(self)} - \beta c B_\theta^{(self)}\right). \tag{52}$$

So we only need the r component of the self-generated electric field and the θ component of the self-generated magnetic field. We will ignore the $B_z^{(self)}$ term in the paraxial limit.

Recall that $V(z) \equiv \phi(0, z)$ is the external potential ϕ on-axis, and to linear order $E_r(r, z) \approx V''(z)r/2$. Similarly, $B(z) \equiv B_z(0, z)$ is the external magnetic field on-axis, and to linear order $B_z(r, z) \approx B(z)$.

Assuming cylindrical symmetry of the beam, the Poisson equation for the self-generated electric potential is:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \phi^{(self)}}{\partial r}\right)+\frac{\partial^2 \phi^{(self)}}{\partial s^2}=-\frac{\rho(r)}{\epsilon_0}. \tag{53}$$

Solving for $\partial\phi^{(self)}/\partial r$ relates the radial electric field to z-derivative of the potential:

$$E_r^{(self)}=-\frac{\partial \phi^{(self)}}{\partial r}=\frac{\lambda(r)}{2\pi\epsilon_0 r}+\frac{1}{2}\frac{\partial^2 \phi^{(self)}}{\partial z^2}r, \tag{54}$$

where λ is the line-charge density,

$$\lambda(r)\equiv\int_0^r 2\pi\rho(\tilde{r})\tilde{r}d\tilde{r}. \tag{55}$$

Above, it is assumed that $\partial^2\phi^{(self)}/\partial z^2$ has no dependence on r (keep lowest-order term).

For the θ component of the self-generated magnetic field, use Ampere's law:

$$\oint \mathbf{B}^{(self)} \cdot d\mathbf{l} = \mu_0 I_{enc} \tag{56}$$

$$2\pi r B_\theta^{(self)} = \mu_0 \int_0^r 2\pi \tilde{r} J_z(\tilde{r}) d\tilde{r} = \mu_0 \lambda(r) v_z \tag{57}$$

$$B_\theta^{(self)} = \frac{\mu_0 \beta c \lambda(r)}{2\pi r} = \frac{\beta}{c} \frac{\lambda(r)}{2\pi \epsilon_0 r}. \tag{58}$$

We can now evaluate $E_r^{(self)} - \beta c B_\theta^{(self)}$:

$$E_r^{(self)} - \beta c B_\theta^{(self)} = \frac{1}{\gamma^2} \frac{\lambda(r)}{2\pi\epsilon_0 r} + \frac{1}{2} \frac{\partial^2 \phi^{(self)}}{\partial z^2} r, \tag{59}$$

We can also repeat the approximation of γ'' with the inclusion of the self-generated electric field.

$$\gamma'' \approx -\frac{q}{mc^2} \frac{\partial^2}{\partial z^2} \left(V + \phi^{(self)} \right). \tag{60}$$

The result is a driving term on the right side of the paraxial ray equation:

$$\boxed{r'' + \frac{(\gamma\beta)'}{(\gamma\beta)} r' + \left(\frac{\gamma''}{2\gamma\beta^2} \right) r + \left(\frac{\omega_c}{2\gamma\beta c} \right)^2 r - \left(\frac{p_\theta}{\gamma m \beta c} \right)^2 \frac{1}{r^3} = \frac{q}{mc^2 \beta^2 \gamma^3} \frac{\lambda(r)}{2\pi\epsilon_0 r}} \tag{61}$$

Outline

1. Paraxial ray equation

2. Envelope equations for axially symmetric beams

3. Envelope equations for elliptically symmetric beams

2.1 Time-evolution of expected value

Let $g(\mathbf{x}, \mathbf{x}')$ be any function of the phase space coordinates. Use brackets to indicate the expected value of g under probability distribution f:

$$\langle g \rangle \equiv \iint g(\mathbf{x}, \mathbf{x}') f(\mathbf{x}, \mathbf{x}', s) d\mathbf{x} d\mathbf{x}'. \tag{62}$$

What is the time-evolution of $\langle g \rangle$?

$$\frac{d\langle g \rangle}{ds} = \frac{d}{ds} \iint g(\mathbf{x}, \mathbf{x}') f(\mathbf{x}, \mathbf{x}', s) d\mathbf{x} d\mathbf{x}' \tag{63}$$

$$
\begin{aligned}
\frac{d\langle g\rangle}{ds} &= \frac{d}{ds}\iint g(\mathbf{x},\mathbf{x}')f(\mathbf{x},\mathbf{x}')d\mathbf{x}d\mathbf{x}' \\
&= \iint g(\mathbf{x},\mathbf{x}')\frac{\partial f(\mathbf{x},\mathbf{x}',s)}{\partial s}d\mathbf{x}d\mathbf{x}' \quad \textit{(Leibniz Rule)} \\
&= \iint g(\mathbf{x},\mathbf{x}')\left[-\mathbf{x}'\cdot\frac{\partial f}{\partial \mathbf{x}} - \mathbf{x}''\cdot\frac{\partial f}{\partial \mathbf{x}'}\right]d\mathbf{x}d\mathbf{x}' \quad \textit{(Vlasov equation).}
\end{aligned} \tag{64}
$$

Since f vanishes at the integration boundary (∞), we can use integration by parts to flip the gradients from f to g:

$$\begin{aligned}\iint \left[\mathbf{x}' \cdot \frac{\partial f}{\partial \mathbf{x}} g\right] d\mathbf{x} d\mathbf{x}' &= -\iint \left[\mathbf{x}' \cdot \frac{\partial g}{\partial \mathbf{x}} f\right] d\mathbf{x} d\mathbf{x}' \\ &= -\langle \mathbf{x}' \cdot \partial g / \partial \mathbf{x} \rangle.\end{aligned} \tag{65}$$

$$\begin{aligned}\iint \left[\mathbf{x}'' \cdot \frac{\partial f}{\partial \mathbf{x}'} g\right] d\mathbf{x} d\mathbf{x}' &= -\iint \left[\mathbf{x}'' \cdot \frac{\partial g}{\partial \mathbf{x}'} f\right] d\mathbf{x} d\mathbf{x}' \\ &= -\langle \mathbf{x}'' \cdot \partial g / \partial \mathbf{x}' \rangle.\end{aligned} \tag{66}$$

Integration by parts (one-dimensional):

$$\int_a^b f(x)g'(x)dx = [f(x)g(x)]_a^b - \int_a^b f'(x)g(x)dx. \tag{67}$$

$$
\begin{aligned}
. &= \iint g(\mathbf{x},\mathbf{x}')\mathbf{x}' \cdot \frac{\partial f(\mathbf{x},\mathbf{x}',s)}{\partial \mathbf{x}} d\mathbf{x}d\mathbf{x}' \\
&= \iiiint g(x,x',y,y') \left[x' \frac{\partial}{\partial x} + y' \frac{\partial}{\partial y}\right] f(x,x',y,y',s) dx dx' dy dy' \\
&= \iiint x' \left[\int g \frac{\partial f}{\partial x} dx\right] dx' dy dy' + \iiint y' \left[\int g \frac{\partial f}{\partial y} dy\right] dx dx' dy' \\
&= -\iiiint f \left[x' \frac{\partial g}{\partial x} + x' \frac{\partial g}{\partial x}\right] dx dx' dy dy' \\
&= -\iint f(\mathbf{x},\mathbf{x}',s) \left[\mathbf{x}' \cdot \frac{\partial g(\mathbf{x},\mathbf{x}')}{\partial \mathbf{x}}\right] d\mathbf{x}d\mathbf{x}' \\
&= -\langle \mathbf{x}' \cdot \frac{\partial g}{\partial \mathbf{x}} \rangle.
\end{aligned} \tag{68}
$$

We can then write $d\langle g\rangle/ds$ as:

$$\frac{d\langle g\rangle}{ds} = \langle \mathbf{x}' \cdot \frac{\partial g}{\partial \mathbf{x}}\rangle + \langle \mathbf{x}'' \cdot \frac{\partial g}{\partial \mathbf{x}'}\rangle. \tag{69}$$

From the definition of the total derivative:

$$\frac{dg}{ds} = \mathbf{x}' \cdot \frac{\partial g}{\partial \mathbf{x}} + \mathbf{x}'' \cdot \frac{\partial g}{\partial \mathbf{x}'}. \tag{70}$$

Therefore, the time-derivative of the expected value is the same as the expected value of the time-derivative:

$$\frac{d\langle g\rangle}{ds} = \langle \frac{dg}{ds}\rangle. \tag{71}$$

To evaluate $d\langle g\rangle/ds$, we can evaluate dg/ds and then perform the averaging. For example:

$$\frac{d}{ds}\langle xx\rangle = 2\langle xx'\rangle \tag{72}$$

$$\frac{d}{ds}\langle x'x'\rangle = 2\langle x'x''\rangle \tag{73}$$

$$\frac{d}{ds}\langle xx'\rangle = \langle xx''\rangle + \langle x'x'\rangle \tag{74}$$

$$\vdots \tag{75}$$

2.2 Evolution of second moments

Define the root-mean-square (rms) radius $\tilde{r} \equiv \sqrt{\langle r^2 \rangle}$.

Evaluate $\tilde{r}'$:

$$\begin{aligned} \frac{d}{ds}\tilde{r}^2 &= \frac{d}{ds}\langle r^2 \rangle \\ \tilde{r}' &= \frac{\langle r r' \rangle}{\tilde{r}}. \end{aligned} \tag{76}$$

Evaluate $\tilde{r}''$:

$$\begin{aligned} \tilde{r}'' &= \frac{\langle r r'' \rangle + \langle r' r' \rangle}{\tilde{r}} - \frac{\langle r r' \rangle \tilde{r}'}{\tilde{r}^2} \\ \tilde{r}'' &= \frac{\langle r r'' \rangle}{\tilde{r}} + \frac{\langle r r \rangle \langle r' r' \rangle - \langle r r' \rangle \langle r r' \rangle}{\tilde{r}^3}. \end{aligned} \tag{77}$$

What is $\langle r r'' \rangle$?

2.3 Evaluation of $\langle rr'' \rangle$

Return to the paraxial ray equation for the radius r (including linear self-fields):

$$r'' + \frac{(\gamma\beta)'}{(\gamma\beta)}r' + \left(\frac{\gamma''}{2\gamma\beta^2}\right)r + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2 r - \left(\frac{p_\theta}{\gamma m\beta c}\right)^2 \frac{1}{r^3} - \frac{q}{mc^2\beta^2\gamma^3}\frac{\lambda(r)}{2\pi\epsilon_0 r} = 0. \tag{78}$$

Rewrite as:

$$r'' + \frac{(\gamma\beta)'}{(\gamma\beta)}r' + \left(\frac{\gamma''}{2\gamma\beta^2}\right)r - \frac{\omega_c\theta' r}{\gamma\beta c} - \theta'^2 r - \frac{q}{mc^2\beta^2\gamma^3}\frac{\lambda(r)}{2\pi\epsilon_0 r} = 0. \tag{79}$$

Multiply by r and take averages:

$$\langle rr'' \rangle + \frac{(\gamma\beta)'}{(\gamma\beta)}\langle rr' \rangle + \left(\frac{\gamma''}{2\gamma\beta^2}\right)\langle r^2 \rangle - \frac{\omega_c}{\gamma\beta c}\langle \theta' r^2 \rangle - \langle \theta'^2 r^2 \rangle - \frac{q}{mc^2\beta^2\gamma^3}\frac{\langle \lambda(r) \rangle}{2\pi\epsilon_0} = 0. \tag{80}$$

$$p_\theta \equiv \gamma m \beta c r^2 \theta' + \frac{m\omega_c r^2}{2} \tag{81}$$

$$\langle p_\theta \rangle = \gamma m \beta c \langle r^2 \theta' \rangle + \frac{m\omega_c \langle r^2 \rangle}{2} \tag{82}$$

$$\vdots \tag{83}$$

$$\frac{\omega_c}{\gamma\beta c}\langle r^2\theta'\rangle = \left(\frac{1}{\gamma\beta m c}\right)^2 \frac{\langle p_\theta\rangle^2}{\tilde{r}^2} - \frac{\langle r^2\theta'\rangle^2}{\tilde{r}^2} - \left(\frac{\omega_c}{2\gamma\beta c}\right)^2 \tilde{r}^2 \tag{84}$$

$$\boxed{\tilde{r}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{r}' + \left(\frac{\gamma''}{2\gamma\beta^2}\right)\tilde{r} + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2 \tilde{r} - \left(\frac{\langle p_\theta\rangle}{\gamma m \beta c}\right)^2 \frac{1}{\tilde{r}^3} - \frac{\varepsilon_r^2}{\tilde{r}^3} - \frac{Q}{2\tilde{r}} = 0} \tag{85}$$

$$Q \equiv \frac{1}{mc^2\beta^2\gamma^3}\frac{q\langle\lambda(r)\rangle}{2\pi\epsilon_0} \tag{86}$$

$$\varepsilon_r \equiv \sqrt{\langle r^2\rangle\langle r'^2\rangle - \langle r r'\rangle^2 + \langle r^2\rangle\langle r^2\theta'^2\rangle - \langle r^2\theta'\rangle^2} \tag{87}$$

Outline

1. Paraxial ray equation

2. Envelope equations for axially symmetric beams

3. Envelope equations for elliptically symmetric beams

Equations of motion ($\mathbf{x} = [x, y]$):

$$\frac{d(\gamma m \dot{x})}{dt} = q\left(E_x + \dot{y}B_z - \dot{z}B_y\right) \tag{88}$$

$$\frac{d(\gamma m \dot{y})}{dt} = q\left(E_y + \dot{z}B_x - \dot{x}B_z\right) \tag{89}$$

Use z as independent variable: $dz = \beta c dt$:

$$x'' + \frac{(\gamma\beta)'}{(\gamma\beta)} x' = \frac{q}{\gamma m \beta^2 c^2} \left[E_x + \beta c \left(y' B_z - B_y\right)\right] \tag{90}$$

$$y'' + \frac{(\gamma\beta)'}{(\gamma\beta)} y' = \frac{q}{\gamma m \beta^2 c^2} \left[E_y + \beta c \left(B_x - x' B_z\right)\right] \tag{91}$$

Add electrostatic potential ϕ in beam rest frame:

$$x'' + \frac{(\gamma\beta)'}{(\gamma\beta)}x' = \frac{q}{\gamma m \beta^2 c^2}\left[E_x + \beta c\left(y' B_z - B_y\right) - \frac{1}{\gamma^2}\frac{\partial \phi}{\partial x}\right] \tag{92}$$

$$y'' + \frac{(\gamma\beta)'}{(\gamma\beta)}y' = \frac{q}{\gamma m \beta^2 c^2}\left[E_y + \beta c\left(B_x - x' B_z\right) - \frac{1}{\gamma^2}\frac{\partial \phi}{\partial y}\right] \tag{93}$$

Assume no external electric field ($E_x = E_y = 0$) and approximate magnetic field as $B_y = Gx$, $B_x = -Gy$, $B_z = 0$. (See Lund lectures for more detail.)

$$x'' + \frac{(\gamma\beta)'}{(\gamma\beta)} x' = +\frac{qG}{\gamma m \beta c} x - \frac{1}{mc^2\beta^2\gamma^3} \frac{\partial \phi}{\partial x} \tag{94}$$

$$y'' + \frac{(\gamma\beta)'}{(\gamma\beta)} y' = -\frac{qG}{\gamma m \beta c} y - \frac{1}{mc^2\beta^2\gamma^3} \frac{\partial \phi}{\partial y} \tag{95}$$

Define rigidity $[B\rho] = p/q = \gamma\beta mc/q$ and linear focusing strengths $\kappa_{x,y} = \pm G/[B\rho]$.

$$x'' + \frac{(\gamma\beta)'}{(\gamma\beta)} x' + \kappa_x(s) x - \frac{q}{mc^2\beta^2\gamma^3}\frac{\partial\phi}{\partial x} = 0 \tag{96}$$

$$y'' + \frac{(\gamma\beta)'}{(\gamma\beta)} y' + \kappa_y(s) y - \frac{q}{mc^2\beta^2\gamma^3}\frac{\partial\phi}{\partial y} = 0 \tag{97}$$

$$\tilde{x} = \sqrt{\langle xx \rangle} \tag{98}$$

$$\tilde{x}' = \frac{\langle xx' \rangle}{\tilde{x}} \tag{99}$$

$$\tilde{x}'' = \frac{\langle xx'' \rangle}{\tilde{x}} + \frac{\langle xx \rangle \langle x'x' \rangle - \langle xx' \rangle \langle xx' \rangle}{\tilde{x}^3} \tag{100}$$

$$= \frac{\langle xx'' \rangle}{\tilde{x}} + \frac{\varepsilon_x^2}{\tilde{x}^3} \tag{101}$$

Calculate $\langle xx'' \rangle$ from equation of motion:

$$x'' + \frac{(\gamma\beta)'}{(\gamma\beta)} x' + \kappa_x(s) x - \frac{q}{mc^2\beta^2\gamma^3} \frac{\partial \phi}{\partial x} = 0 \qquad (102)$$

$$x x'' + \frac{(\gamma\beta)'}{(\gamma\beta)} x x' + \kappa_x(s) x^2 - \frac{q}{mc^2\beta^2\gamma^3} x \frac{\partial \phi}{\partial x} = 0 \qquad (103)$$

$$\langle x x'' \rangle + \frac{(\gamma\beta)'}{(\gamma\beta)} \langle x x' \rangle + \kappa_x(s) \langle x^2 \rangle - \frac{q}{mc^2\beta^2\gamma^3} \langle x \frac{\partial \phi}{\partial x} \rangle = 0 \qquad (104)$$

$$\tilde{x}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{x}' + \kappa_x(s)\tilde{x} - \frac{\varepsilon_x^2}{\tilde{x}^3} - \frac{q}{mc^2\beta^2\gamma^3}\frac{\langle x\partial\phi/\partial x\rangle}{\tilde{x}} = 0 \tag{105}$$

$$\tilde{y}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{y}' + \kappa_y(s)\tilde{y} - \frac{\varepsilon_y^2}{\tilde{y}^3} - \frac{q}{mc^2\beta^2\gamma^3}\frac{\langle y\partial\phi/\partial y\rangle}{\tilde{y}} = 0 \tag{106}$$

What is $\langle x \frac{\partial \phi}{\partial x} \rangle$?

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial x^2} \right) \phi(x, y) = -\frac{\rho(x, y)}{\epsilon_0}. \tag{107}$$

Assume elliptical symmetry: $\rho(x, y) = \rho\left((x/\tilde{x})^2 + (y/\tilde{y})^2 \right)$. It can then be shown that:

$$\langle x \frac{\partial \phi}{\partial x} \rangle = -\frac{\lambda}{4\pi\epsilon_0} \frac{\tilde{x}}{\tilde{x} + \tilde{y}}, \tag{108}$$

$$\langle y \frac{\partial \phi}{\partial y} \rangle = -\frac{\lambda}{4\pi\epsilon_0} \frac{\tilde{y}}{\tilde{x} + \tilde{y}}. \tag{109}$$

$$\tilde{x}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{x}' + \kappa_x(s)\tilde{x} - \frac{\varepsilon_x(s)^2}{\tilde{x}^3} - \frac{Q}{2\,(\tilde{x}+\tilde{y})} = 0, \tag{110}$$

$$\tilde{y}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{y}' + \kappa_y(s)\tilde{y} - \frac{\varepsilon_y(s)^2}{\tilde{y}^3} - \frac{Q}{2\,(\tilde{x}+\tilde{y})} = 0, \tag{111}$$

$$Q = \frac{\lambda}{mc^2\beta^2\gamma^3}\frac{q}{2\pi\epsilon_0}. \tag{112}$$

- These are the transverse rms envelope equations for an elliptically symmetric distribution in linear (magnetic) external fields. Acceleration is included, but no energy spread.
- Does not assume uniform density! Proved by Sacherer.
- Note: space charge kick is always the same in both planes!

3.1 References